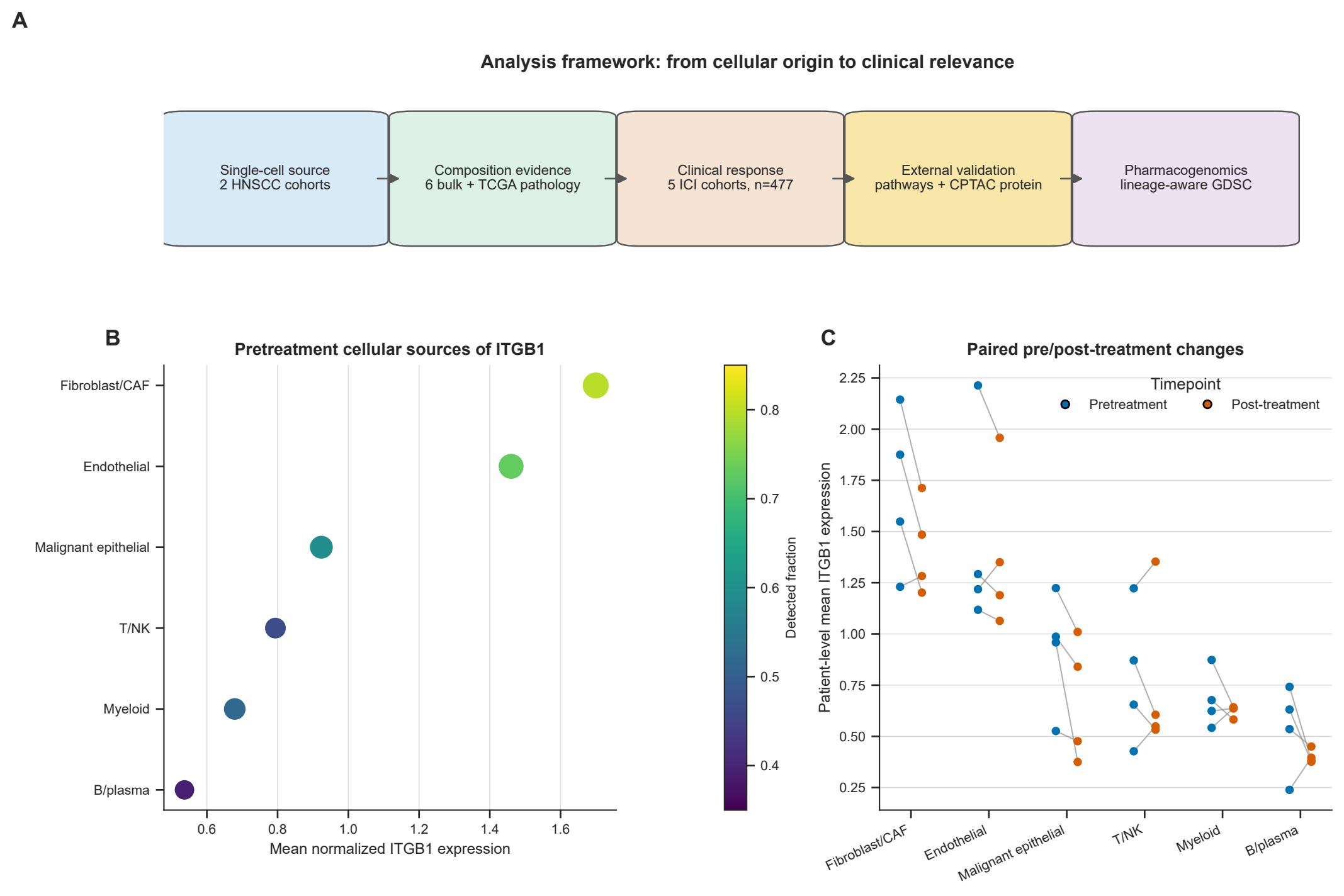

A
Analysis framework: from cellular origin to clinical relevance
Single-cell source
2 HNSCC cohorts
Composition evidence
6 bulk + TCGA pathology
Clinical response
5 ICI cohorts, n=477
External validation
pathways + CPTAC protein
Pharmacogenomics
lineage-aware GDSC
B
Pretreatment cellular sources of ITGB1
Fibroblast/CAF
Endothelial
Malignant epithelial
T/NK
Myeloid
B/plasma
0.6
0.8
1.0
1.2
1.4
1.6
Mean normalized ITGB1 expression
Detected fraction
0.8
0.7
0.6
0.5
0.4
C
Paired pre/post-treatment changes
Timepoint
Pretreatment
Post-treatment
Patient-level mean ITGB1 expression
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
Fibroblast/CAF
Endothelial
Malignant epithelial
T/NK
Myeloid
B/plasma